

# Scalars, vectors, contact and non-contact forces

**Forces**  
Paper 2 · Combined and Separate  
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                            | ANSWER                                                                  | ✓                        |
|----|---------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------|
| 1  | Define a scalar quantity.                                           | A quantity that has magnitude only.                                     | <input type="checkbox"/> |
| 2  | Give three examples of scalar quantities.                           | Any three from: energy, time, mass, distance, speed, temperature        | <input type="checkbox"/> |
| 3  | Define a vector quantity.                                           | A quantity that has both magnitude and direction.                       | <input type="checkbox"/> |
| 4  | Give three examples of vector quantities.                           | Any three from: force, displacement, velocity, acceleration, momentum   | <input type="checkbox"/> |
| 5  | State how a vector is represented on a diagram.                     | By an arrow.                                                            | <input type="checkbox"/> |
| 6  | Explain why an arrow is a suitable way to represent a vector.       | Its length shows the magnitude and its direction shows the direction.   | <input type="checkbox"/> |
| 7  | State what a force is.                                              | A push or pull on an object due to an interaction with another object.  | <input type="checkbox"/> |
| 8  | Define a contact force.                                             | A force between objects that are physically touching.                   | <input type="checkbox"/> |
| 9  | Give three examples of contact forces.                              | Any three from: friction, air resistance, tension, normal contact force | <input type="checkbox"/> |
| 10 | Define a non-contact force.                                         | A force between objects that are physically separated.                  | <input type="checkbox"/> |
| 11 | Give three examples of non-contact forces.                          | Gravitational force (weight), electrostatic force, magnetic force       | <input type="checkbox"/> |
| 12 | When two objects interact, state which of them experiences a force. | Both: the forces are equal in size and opposite in direction.           | <input type="checkbox"/> |